

Da jia hao (따자하오 / 안녕하세요)

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Fāng'àn Kāifā Tuándùì. (무치엔 푸즈어 산씽 띠엔즈으 춘추치이 쉬이에뿌 지에쥬에
팡안 카이푸 투안뚜이 / 현재 삼성전자 솔루션 개발팀장을 맡고 있습니다)

It is a pleasure to be here with you at OCTS.

Today, I would like to share our perspective
on next-generation flash storage technologies
designed to address the explosive growth of context
in the era of Agentic AI.

What does a factory do?

It takes raw materials in and produces products at scale. That is exactly what is happening in AI industry today.

The industry has shifted—from training individual models, to **mass-producing intelligence at scale**. We are now entering what we call the **AI Token Factory era**.

And the market evolution clearly reflects this transformation. It is projected to grow from \$318 billion in 2025 to nearly one trillion dollars by 2029.

At the same time, workloads are fundamentally changing. We are moving from traditional LLMs to Agentic AI, where multiple agents maintain memory, reason across steps, and operate continuously across sessions. This is creating incredible opportunity—but also a new bottleneck. And surprisingly, it is not at a compute process.

Inside this factory, a Context Explosion is unfolding. At the center of it is the KV cache, the mechanism that enables models to remember previous tokens. And this KV cache is scaling explosively across three dimensions: **context length, concurrency, and session lifetime**. Even with aggressive optimizations—such as DeepSeek V4 Pro reducing KV cache usage by up to 90%—the scale continues to grow. A single user can still generate around 10GB of KV cache, and at just one thousand concurrent users, this quickly expands to 10TB at system level.

But Agentic AI takes this even further. Unlike traditional LLMs, agents do not reset after each interaction. They maintain persistent sessions, store tool outputs, retain prior conversations, and accumulate reasoning traces over time. Memory is no longer transient—it becomes continuously **accumulated**. And as a result, KV cache demand is rapidly exceeding the limits of traditional memory systems. Keeping this entirely in HBM and DRAM is no longer enough. The KV cache must **move beyond main memory**.

As context windows scale toward millions of tokens, KV cache becomes one of the largest system-level memory consumers.

And today's memory hierarchy—HBM and DRAM—is **no longer sufficient**.

At the same time, AI is evolving into multi-agent systems, where context must be shared and synchronized in real time.

This leads to one clear conclusion.

Now, we are redefining how storage participates in the process of AI inference.

Within SSD-based KV cache storage, we establish two optimized domains:

TLC-based storage for Active KV Cache, built for high-performance, real-time inference.

And QLC-based storage for Archive KV Cache, built for massive-scale, long-term contextual retention.

Together, they form a unified KV cache storage architecture, balancing performance and scale without compromise.

So what does the storage for KV cache require?

It requires bandwidth, capacity, cost efficiency, and persistence—all at once. Now let's look at the options.

HBM is extremely fast, but limited in capacity and extremely expensive.

DRAM offers more capacity, but still constrained in density and cost.

HDD provides scale, but latency makes it unusable for real-time inference.

That leaves one viable answer. **SSD** is the optimized tier that **satisfies all requirements** together.

For Active KV Cache Storage, we need two things: maximum performance and massive capacity.

To address this, let me introduce PM1763.

Built on Samsung **Foundry's 4nm** controller, it pushes PCIe Gen6 performance to its limits. It delivers high efficiency, supports liquid cooling environments, and comes in flexible form factors such as E1.S and E3.S, enabling scalable deployment across AI systems.

It is already validated by key customers and **moving toward mass production.**

As AI workloads scale, rack density is increasing rapidly, and traditional air cooling is reaching physical limits.

The industry is now shifting toward liquid cooling, which removes heat **directly at the source** and enables significantly **higher system density and performance**.

In this new environment, storage can no longer remain thermally isolated.

It must operate reliably under the same high-density, high-thermal conditions as compute.

That is why liquid-cooled infrastructure–ready storage is becoming essential.

Our solution is designed to be fully compatible with **direct liquid cooling environments**, ensuring sustained performance, improved reliability, and stable operation in next-generation AI data centers.

We already have a strong answer for today with PM1763.

But the demand is not slowing down—
it is accelerating.

Every generation of AI pushes performance boundaries further.

And we are ready for what comes next.

Samsung is already preparing for **PCIe Gen7**.

Our next architecture is being designed to once again hit
interface limits at scale.

So when the industry takes the next leap,

we will be ready.

Now let's move to the storage for **Archive KV Cache**.
As AI systems scale, not all context disappears after use. A large portion becomes reusable intelligence—past sessions, reasoning traces, and tool outputs that still hold value. This data continues to grow without limit. To handle this, we need **ultra-scale SSD storage capacity**. That's what we have prepared for, a high capacity SSD solution, built to absorb this ever-growing archive KV data. This transforms past context into a **reusable** asset, reducing **recomputation** and **improving efficiency** across the system.

Now, let me introduce Samsung's storage solution for Archive KV Cache: **BM1773**.

This is a **high-capacity SSD** designed for **extreme-scale contextual storage**.

It delivers up to 256 terabytes in a single drive, the highest in the industry. And as the **only vendor** offering E3.S at 256TB, Samsung enables petabyte-scale storage within a single rack, achieving approximately **35% better area efficiency** compared to the others.

Beyond capacity,
BM1773 is engineered for real-world AI workloads.
It provides sufficient performance
to efficiently store and retrieve KV cache data,
leveraging a Gen5 interface architecture optimized for large-
scale data movement.
And with full liquid cooling compatibility,
it is well-suited for high-density,
next-generation AI data center deployments.

With these solutions, we complete a unified architecture for KV cache lifecycle management.

From real-time inference to long-term contextual retention, KV data can now be managed efficiently across tiers.

Ultimately, this is not just about storage.

It is about **enabling AI systems** to scale beyond today's limitations—making intelligence **more continuous, more efficient, and more accessible**.

With this foundation, AI is moving toward everyday ubiquity. And **Samsung will be at the core** of that transformation.

Thank you.